Graph Theoretic Clustering

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Renyi entropy

- Free parameter: α or q
- Denoted as $H_{R\alpha}$ or H_{Rq}

$$H_\alpha(X) = \frac{1}{1-\alpha} \log_2 \left(\sum_{i=1}^{n} p_i^\alpha \right)$$

H_{R2} (i.e. $\alpha = 2$)

- $H_{R2} = -\log \sum_i p_i^2$

Information potential

$$H_\alpha(X) = \frac{1}{1-\alpha}\log\left(\sum_{k=1}^{N} p_k^\alpha\right) = -\log\left(\sum_{k=1}^{N} p_k^\alpha\right)^{\frac{1}{\alpha-1}}$$

$$Information\ Potential = V^\alpha(x) = \sum_{k=1}^{N} p_k^\alpha$$

Information potential is
Sum of powered probabilities

Information potential is Sum of powered probabilities

- How to get PDF?
- Parzen windowing

Example

Given a set of five data points x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6

Find **<u>Parzen probability density function</u>** (pdf) estimates at x = 3, using the Gaussian function with σ = 1 as window function

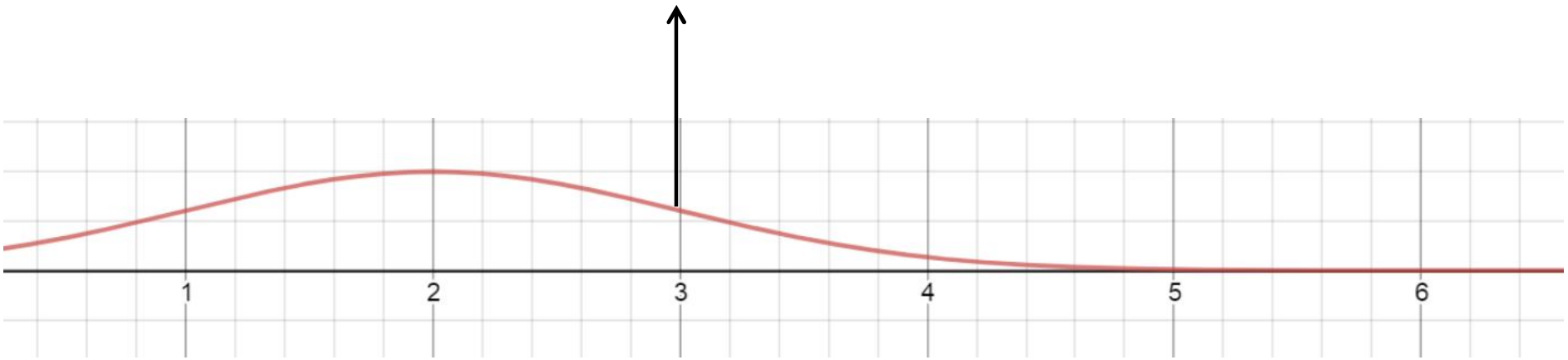
Algorithm

<u>x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6</u>

1. Place a Gaussian at x=2 i.e. μ = 2
2. Find its value @ x=3
3. Place a Gaussian at x=2.5 i.e. μ = 2.5
4. Find its value @ x=3
5. ..
6. ..
7. ..
8. ..
9. Place a Gaussian at x=6 i.e. μ = 6
10. Find its value @ x=3

Gaussian with μ = 2

Find value x=3

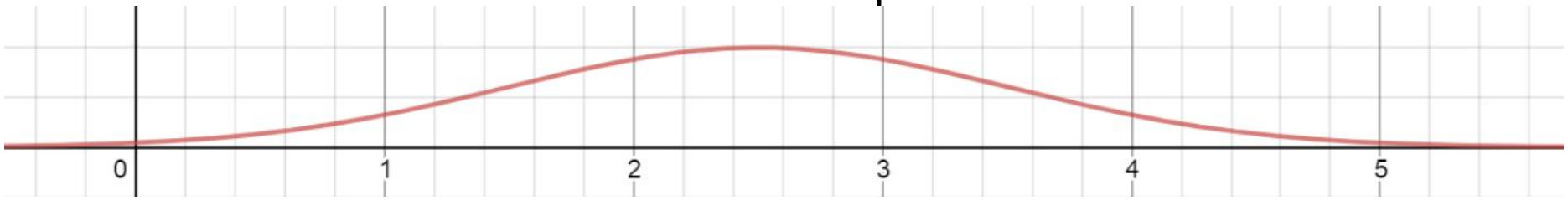


$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_1 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2-3)^2}{2}\right) = 0.2420$$

Gaussian with μ = 2.5

Find value x=3

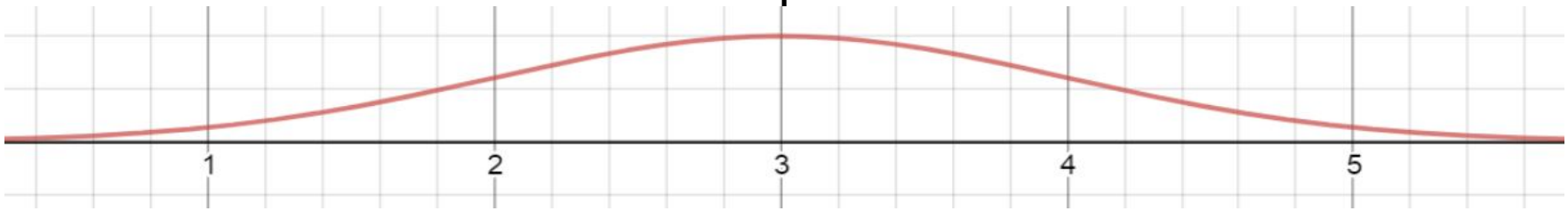


$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_2 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2.5 - 3)^2}{2}\right) = 0.3521$$

Gaussian with μ = 3

Find value x=3 (we get maximum value)

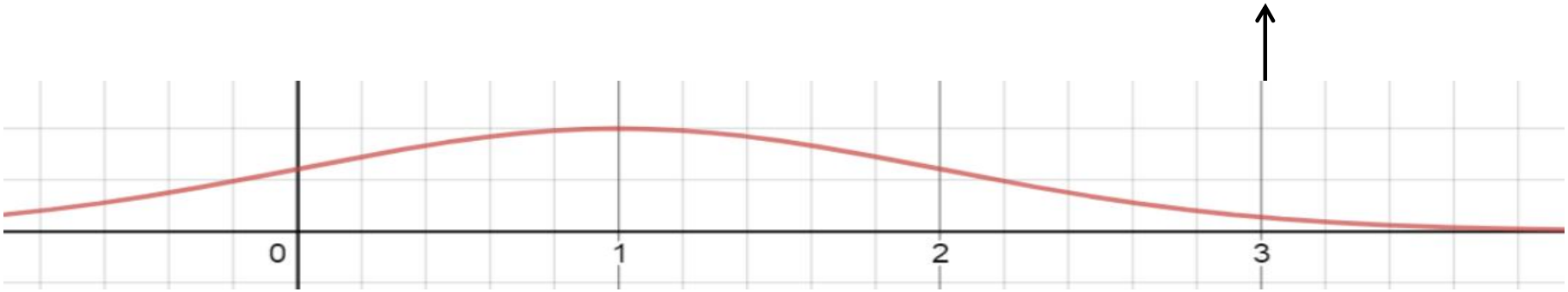


$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_3 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(3 - 3)^2}{2}\right) = 0.3989$$

Gaussian with μ = 1

Find value x=3

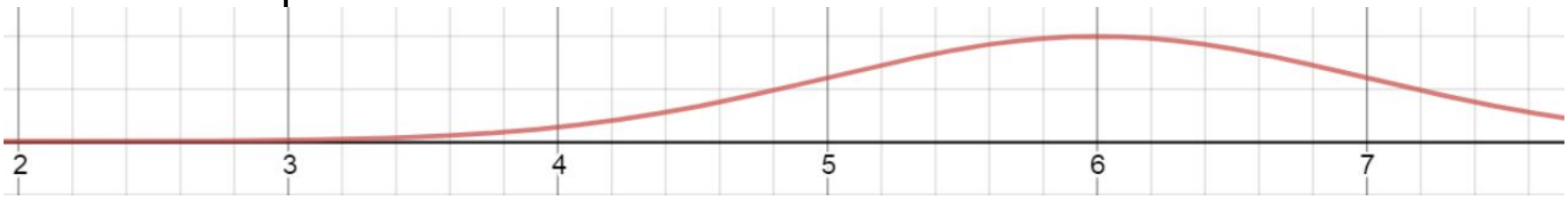


$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_4 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(1 - 3)^2}{2}\right) = 0.054$$

Gaussian with μ = 6

Find value x=3



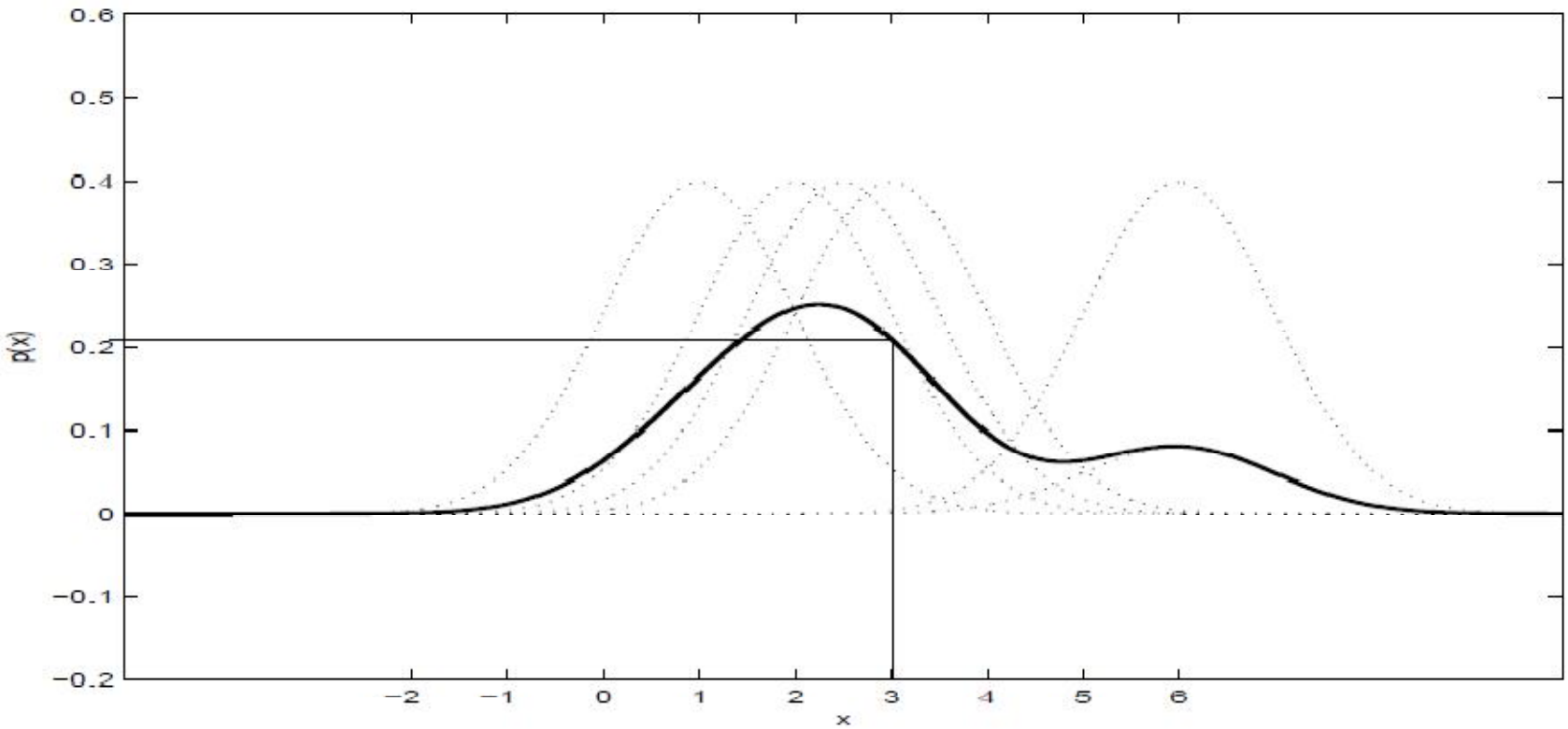
$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_5 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(6 - 3)^2}{2}\right) = 0.0044$$

$$p(x = 3) = (0.2420 + 0.3521 + 0.3989$$

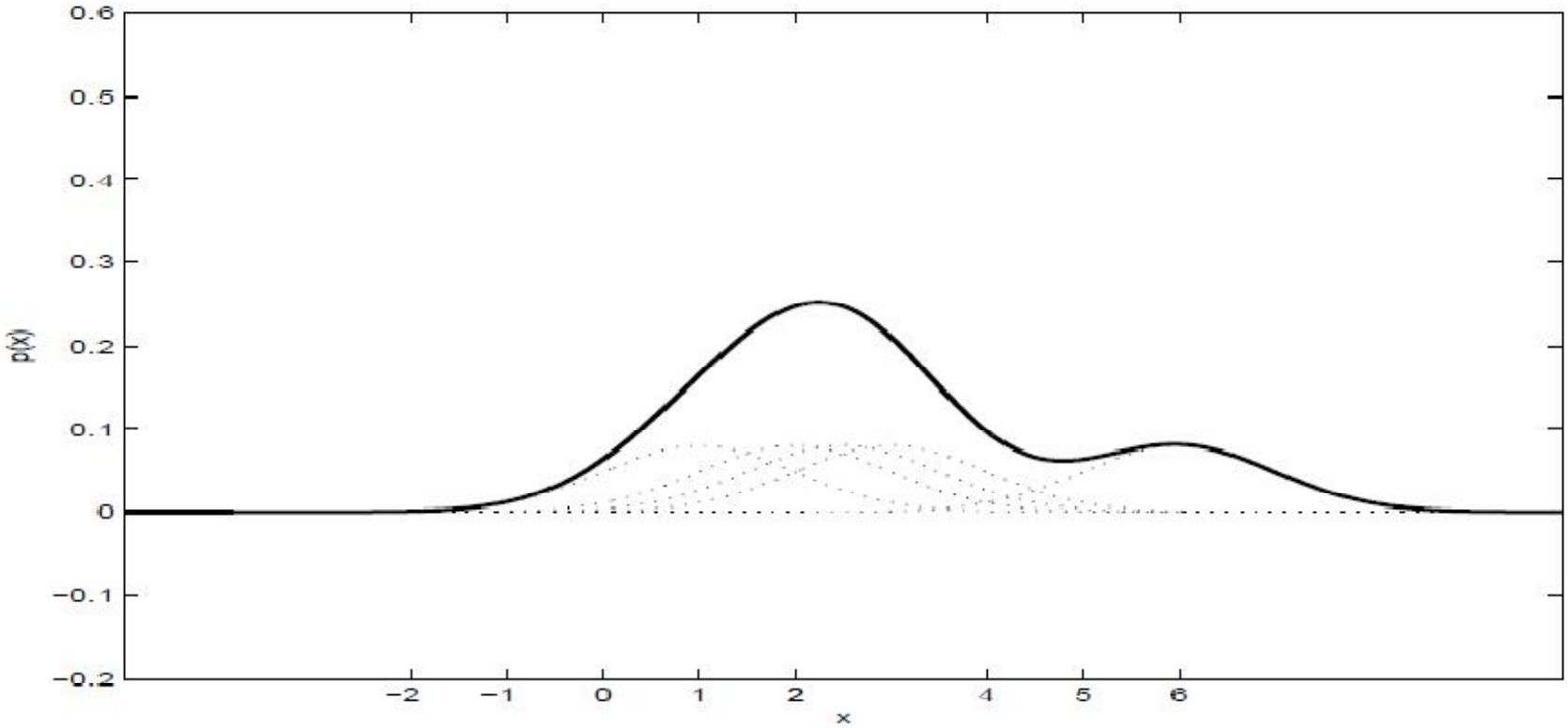
$$+0.0540 + 0.0044)/5 \quad = 0.2103$$

x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6



What is p(3)? Answer is 0.21

Given: x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6



What is p(x) in general?

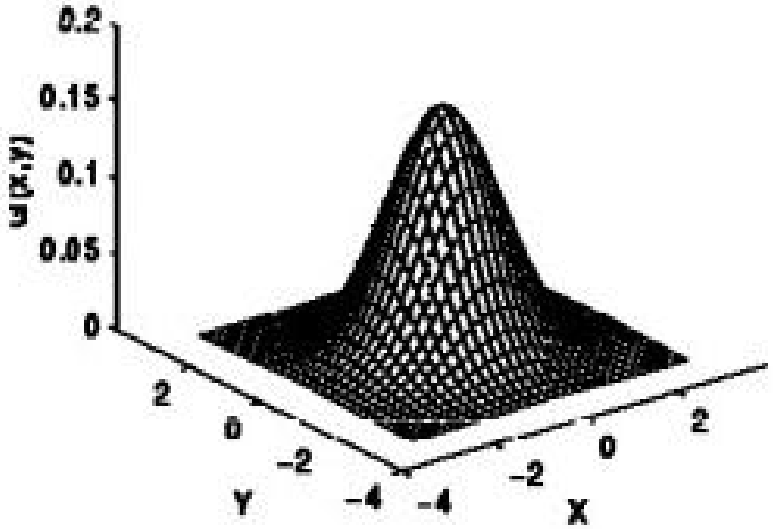
Answer is the estimated curve

If data is two dimensional ...[1]

- Use two dimensional Gaussian function to estimate PDF

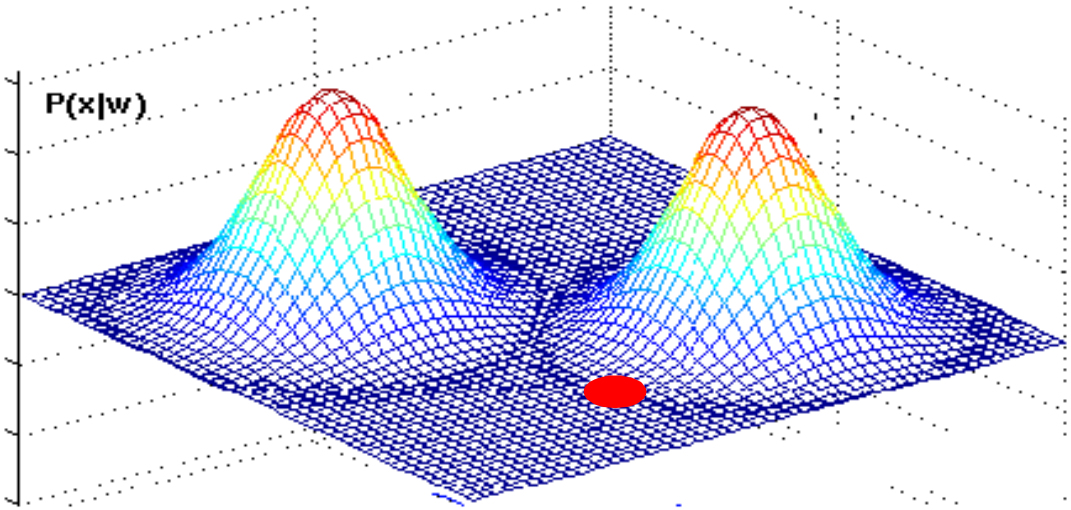
$$G(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

A graphical representation of the 2D Gaussian distribution with mean(0,0) and σ = 1



If data is two dimensional ...[2]

- Place the Gaussian @ various points
- Find their contributions @ the given point
- Sum up all the contributions
- Take average



- Assume we have 3 points on a plane
- @ red color point the PDF is given by average contributions from other two Gaussians

Parzen Window

- Number of data points (N)

<u>To calculate the density @ one point</u>

- N Gaussian computations and then summing up
  - One Σ

Information potential (IP)

- Need to know PDf @ N points and then summing up
  - One more Σ
  - i.e. Two Σ

$$Information\ Potential = V^{\alpha}(x) = \sum_{k=1}^{N} p_k^{\alpha}$$

- Quadratic information potential with Gaussian kernel

$$\left(\frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} G_{\sigma\sqrt{2}}(x_j - x_i) \right)$$

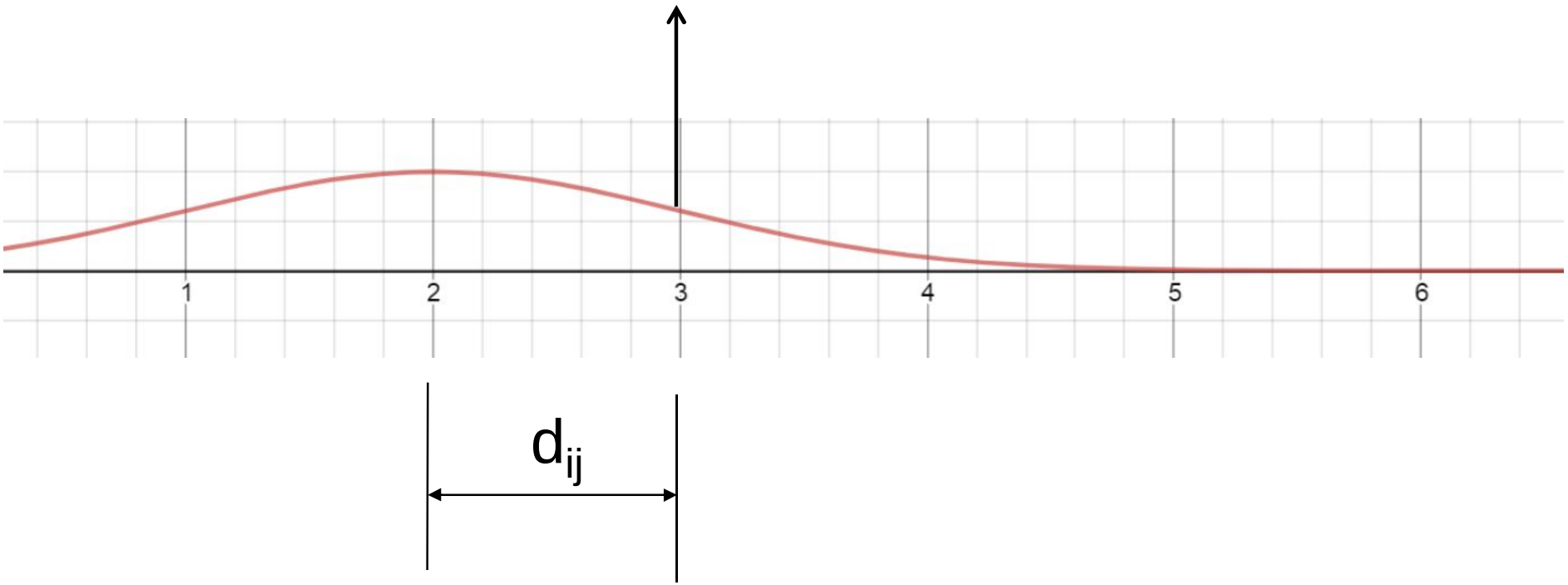
$$\hat{V}(X) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \hat{V}_{i,j}$$

$$d_{i,j} = x_i - x_j$$

$$V_{ij} = G(d_{ij})$$

What is $G(d_{ij})$?

Find value x=3



Information force

Information force on sample xj due to sample xi

$$\hat{F}(i) = \frac{-1}{2N\sigma^2} \sum_{j=1}^{N} \boxed{\hat{V}_{i,j} d_{i,j}}$$

where

$$d_{i,j} = x_i - x_j$$

$$V_{ij} = G(d_{ij})$$

Total information force acting on sample

Information force and information potential

$$\hat{F}(i) = \frac{-1}{2N\sigma^2} \sum_{j=1}^{N} \hat{V}_{i,j} d_{i,j}$$

where

$$d_{i,j} = x_i - x_j$$

$$V_{ij} = G(d_{ij})$$

$$\hat{V}(X) = \frac{1}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} \hat{V}_{i,j}$$

$G(d_{ij})$

- $G(d_{ij})$ – Gaussian placed @ x_j point, and evaluated @ x_i point
- $G(d_{ij})$ is always positive
- Both IP and IF are functions of $G(d_{ij})$

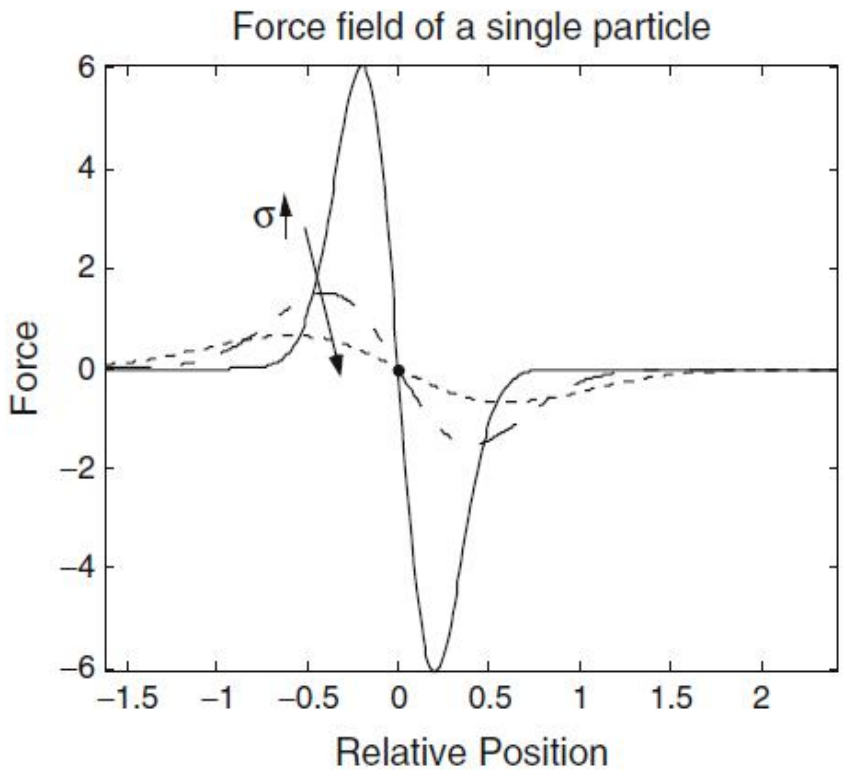
d_{ij}

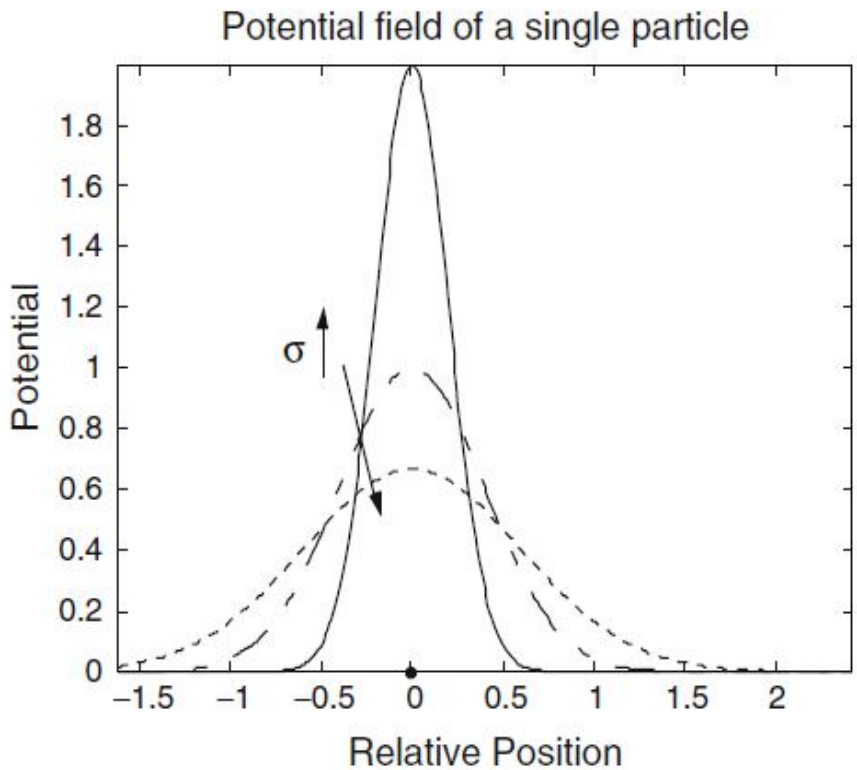
- $d_{ij} = x_i - x_j$
- Can be positive or negative
- IF = $G(d_{ij})$ x d_{ij}
- IF could be +ve or -ve

IP and IF

- IF +ve and –ve
- IF depends on both $G(d_{ij})$ & d_{ij}

- IP always +ve
- IP depends on $G(d_{ij})$





IF for two dimensional

Dimension X

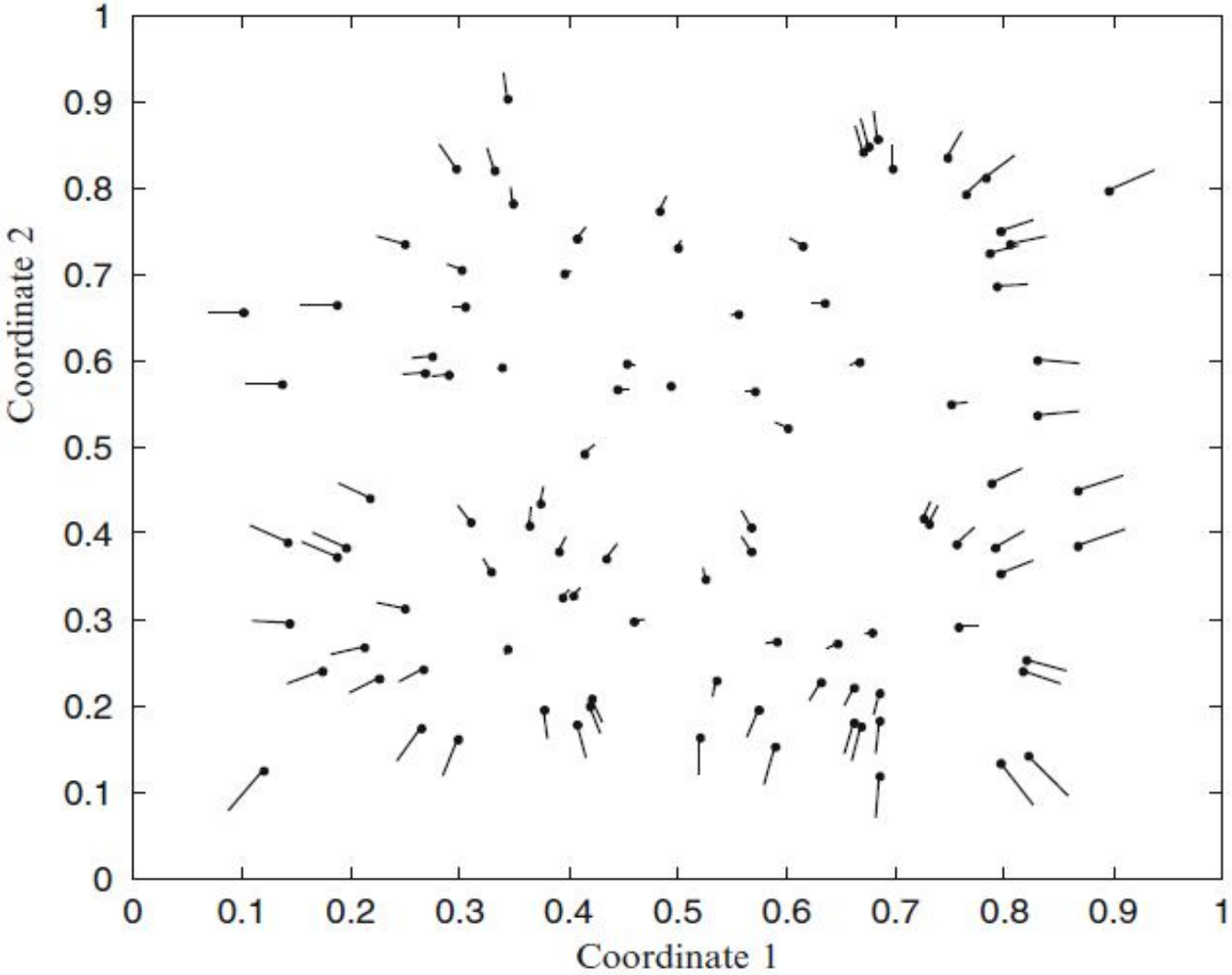
- $d_{ij} = x_i - x_j$
- $IF_x = G(d_{ij}) \times d_{ij}$

Dimension Y

- $d_{ij} = y_i - y_j$
- $IF_y = G(d_{ij}) \times d_{ij}$

- Combine IF_x and IF_y
- Vector sum
- Produces magnitude as well as direction

Quadratic information forces on 2 featured data points

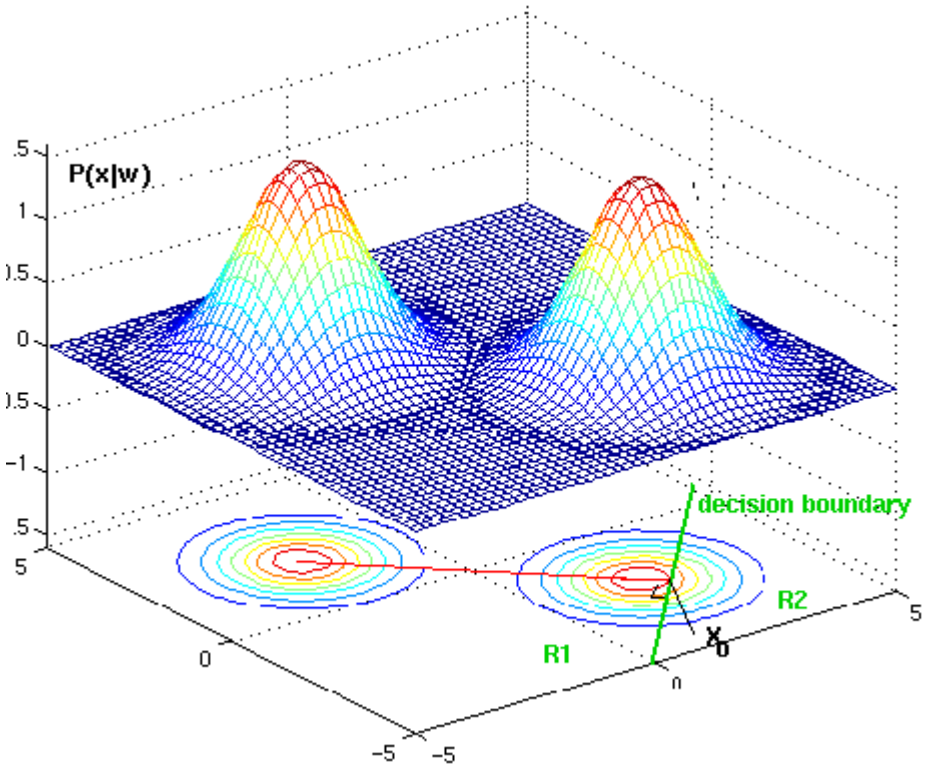


Information forces for clustering

- Some points become center of cluster
- IF of many surrounding points oriented towards them

Directed tree

- Create graph out of N nodes
- Use their IF as cost
- More the IF better the connection

General constraints

- $d_{ij} < 3\sigma$
- $|F| > 0$